

Math 242, S16
Quiz 5

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Given the following initial value problem:

$$\frac{dP}{dt} = 15P - 3P^2, \quad P(0) = 3.$$

(a) Solve the initial value problem for $P(t)$.

$$\begin{aligned} \frac{dP}{dt} &= 3P(5-P) & P(0) &= 3 \\ \frac{1}{P(5-P)} dP &= 3 dt & \Rightarrow \frac{3}{5-3} &= B e^{15 \cdot 0} \Rightarrow B = \frac{3}{2} \\ \int \frac{1}{P(5-P)} dP &= \int 3 dt & \text{thus} & \\ \frac{1}{5} \int \frac{1}{P} + \frac{1}{5-P} dP &= 3t + C, & \frac{P}{5-P} &= \frac{3}{2} e^{15t} \\ \ln|P| - \ln|5-P| &= 15t + C & \Rightarrow P(t) &= \frac{15}{3+2e^{-15t}} \\ \ln\left|\frac{P}{5-P}\right| &= 15t + C \\ \left|\frac{P}{5-P}\right| &= e^{15t+C} \\ \frac{P}{5-P} &= B e^{15t} \end{aligned}$$

(b) Find the limit $M = \lim_{t \rightarrow +\infty} P(t)$.

$$\lim_{t \rightarrow +\infty} P(t) = \lim_{t \rightarrow +\infty} \frac{15}{3+2e^{-15t}} = \frac{15}{3} = 5$$